

used in the charging device increase the complexity and cost

- Most of the mobile devices, when connected to a cabled charging system, can be moved around (in a limited range) and can be operated while charging. Whereas, in most of the static wireless charging systems, the mobile device must be left on a pad to charge and cannot be moved around or easily operated while charging

Standards for wireless charging

Several organisations worldwide formulate standards for inductive charging. The Wireless Power Consortium (WPC) established in 2008 created the Qi (pronounced “chee”) standards in 2010, which became the most popular standards—covering inductive charging and short-distance (1.5cm or less) electromagnetic charging.

Japan established Broadband Wireless Forum (BWF) in 2009 and the Energy Harvesting Consortium (EHC) was founded in Japan in 2010. Korea established the Korean Wireless Power Forum (KWPF) in 2011. In 2012, the Institute of Electric and Electronic Engineers (IEEE) initiated the formation of the Alliance for Wireless Power (A4WP) and the Power Matter Alliance (PMA). The Wireless Power Consortium for Practical Applications (WiPoT) was formed in 2013.

In 2018, The Qi Wireless Standard was adopted for use in military equipment in North Korea, Russia, and Germany. The WPC, a 296-member consortium, included Apple, Google, Verizon and several other leading electronics manufacturers.

In 2014, the PMA formulated inductive charging specifications—Powermat for the technology used for wireless charging in coffee shops and airports. The Starbucks started using Powermat in 2015 for their nearly 8,000 coffee shops.

The A4WP and the PMA joined together to form the AirFuel Alliance (AFA), which now has 110 members including Dell, Duracell, Samsung, and Qualcomm. Duracell has more



Prof. Herbert Kohler from Daimler who predicts wireless mass availability after 2026 (Credit: <https://media.daimler.com>)

than 1,500 charging spots in the US and 1,000 charging spots in European airports, hotels, and cafes using AFA standards.

The two main standards, Qi and AFA, are quite similar but use different transmission frequencies and connection protocols. Hence, the devices compatible with one standard are not necessarily compatible with the other standard. However, devices compatible with both the standards are available.

An emerging standard is SAE J2954, which covers inductive car charging up to 11kW. ISO 15118 for Vehicle to Grid communication is an internationally adopted standard issued recently by the International Organization for Standards (ISO), Geneva.

Conclusion

High-performance, safe, and cost-effective EV wireless charging has the potential to revolutionise road transportation. The capacitive and inductive charging systems both offer enormous opportunities for research, especially in designing high-frequency power electronics and near-field couplers. Some of the significant areas needing further research are the effects of long-term human exposure to electric and magnetic fields; methods to detect living and foreign objects in the proximity of the systems; optimal charging power levels and air spacing for cost-effectiveness; cost-effective technology for embed charging roadways; and impact of system deployment on the electric

grids. Research and experimentation are currently underway in these fields.

High initial costs associated with installing and using the infrastructure for dynamic wireless charging systems are a big hurdle, more so as this would benefit only EVs, a small percentage of the total vehicles currently on the road. Another problem is the methodology to be adopted for computing the energy consumption of each vehicle while charging from the lane.

The other related issues needing attention are: eliminating or controlling safety hazards associated with the large amounts of heat and other electromagnetic radiations produced between the charging surfaces. For example, the safety hazard faced by a person with an implanted pacemaker travelling inside the EV. Another downside is the need for precise alignment between the inductive pick up and the charging facility.

A beginning has been made and wireless charging systems are already available commercially. Smartphone manufacturers like Apple and Samsung have already started adding this technology into their high-end devices. The wireless charging technology developed for charging of electric vehicles has potential for very wide application in wirelessly powered biomedical implants, humanoid robots, and supersonic hyperloop transport.

The technological challenges in these fields are exciting and the possibilities endless, as numerous problems related to performance, cost, and safety need to be overcome before the vision of wirelessly powered EVs can be finally realised. Professor Herbert Kohler, Daimler’s Future Mobility Head, has predicted that the wireless charging for EVs may become available for common use after 2026 only, when all the associated aspects have been investigated. **EFY**



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